

Lab 5: Noise and Probability in Imaging

Due Wednesday, March 4, 11:59pm

In this lab, we explore one of the most fundamental limitations of imaging systems: noise. No matter how advanced a camera is, every captured image is affected by randomness arising from both incoming light and sensor electronics. Understanding where this noise comes from and how it behaves is essential for designing, analyzing, and improving imaging systems.

Rather than treating noise as an abstract concept, you will study it through probability models and direct simulation. You will begin by reviewing key probability distributions that appear in imaging, including the Poisson and Gaussian distributions. Using these models, you will then simulate photon noise and sensor noise and observe how they affect images under different lighting and exposure conditions.

By the end of this lab, you should be able to explain how different sources of noise arise in digital cameras, how they depend on scene brightness and exposure, and how they influence image quality and signal-to-noise ratio.

1 Probability Review

Problem 1.1: For each random variable, state whether it is discrete or continuous, identify an appropriate probability distribution, and briefly explain your reasoning.

- (a) Number of photons arriving at a pixel in 1 ms

Discrete, Poisson $[+\infty]$

Photons arrive as individual particles independently sampled

- (b) A random number generated uniformly in $[0, 1]$

Continuous, Unif $[0, 1]$

This random number generator returns real numbers with equal probability

- (c) Voltage noise measured at a camera sensor

Continuous, Normal $(0, \sigma_{\text{read}})$

Voltage can take any real value, which can be modeled with a continuous gaussian.

- (d) The number of electrons generated by dark current.

Discrete, Poisson $[+N]$

Electrons are counted individually like in 1a

Problem 1.2: During a single exposure, each pixel records a photon count at a discrete sample index n . Assume photon arrivals are independent random events, and we measure the total count in a fixed time window. Let $x[n]$ be the number of photons detected at index n , and suppose the expected count is $\mathbb{E}[x[n]] = 50$.

- (a) Write the probability mass function (PMF) of $x[n]$.

$$\frac{50^k e^{-50}}{k!}$$

- (b) Compute
- $\mathbb{E}[x[n]]$
- and
- $\text{Var}(x[n])$
- .

$$\mathbb{E}[x[n]] = 50, \quad \text{Var}(x[n]) = 50$$

- (c) If the exposure time doubles (and the scene brightness stays the same), what happens to the mean, variance, and the signal-to-noise ratio (SNR)?

The mean and variance both double

SNR increases by a factor of $\sqrt{2}$

Problem 1.3: Electronic read noise is modeled as a zero-mean random sequence $h[n]$, where each sample satisfies

$$h[n] \sim \mathcal{N}(0, 4).$$

This type of distribution commonly arises from the sum of many small independent electronic fluctuations.

- (a) What is the standard deviation of
- $h[n]$
- ?

$$\sigma = 2$$

- (b) What fraction of values lie within
- $\pm 2\sigma$
- ?

$$95\%$$

- (c) If we amplify the measured signal by a factor of 3, what happens to the noise variance?

Variance would be amplified by a factor of 9

$$\text{Var}(3h[n]) = 3^2 \cdot 4 = 36$$

Problem 1.4: A simplified discrete sensor output model is

$$y[n] = x[n] + h[n],$$

where

$$x[n] \sim \text{Poisson}(\lambda), \quad h[n] \sim \mathcal{N}(0, \sigma_h^2).$$

Assume $x[n]$ and $h[n]$ are independent.

In this problem, $\lambda = 80$ and $\sigma_h^2 = 9$ (so $h[n] \sim \mathcal{N}(0, 9)$).

- (a) Compute
- $\mathbb{E}[y[n]]$
- and
- $\text{Var}(y[n])$
- .

$$\mathbb{E}[y[n]] = \mathbb{E}[x[n]] + \mathbb{E}[h[n]] = 80 + 0 = 80$$

$$\text{Var}[y[n]] = \text{Var}(x[n]) + \text{Var}(h[n]) = 80 + 9 = 89$$

- (b) Which noise source dominates: the shot noise
- $x[n]$
- or the read noise
- $h[n]$
- ?

The shot noise dominates, since $\text{Var}(x[n]) \gg \text{Var}(h[n])$

- (c) If the scene brightness doubles, how do
- $\mathbb{E}[y[n]]$
- and
- $\text{Var}(y[n])$
- change?

$$\mathbb{E}[y[n]] = 160$$

$$(\lambda \rightarrow 2\lambda)$$

$$\text{Var}[y[n]] = 169$$

2 Measuring Noise Using the Canon R10

In this section, you will use a real digital camera to experimentally measure and analyze image noise using repeated captures of a static scene. You will verify the noise models using real data.

Problem 2.1: Data Collection

1. Set up the camera in a static scene and keep it fixed during capture.
2. Use a static scene with uniform lighting (e.g. blank wall or screen).
3. Set the camera to : **Manual mode, Fixed aperture, Fixed exposure time, RAW format**
4. Capture 30 images each at different ISO settings. (e.g. ISO 800, ISO 1600, and ISO 3200)

Problem 2.2: For each ISO folder, you are given N repeated RAW captures of the same static scene.

The starter code already computes:

- a **mean image** (pixelwise mean across the N captures), and
- a **variance image** (pixelwise sample variance across the N captures).

Your task:

1. Divide the mean image and variance image into **non-overlapping** $p \times p$ patches (use $p = 32$).
2. For each patch, compute two values:
 - patch mean μ : the average of pixels in the **mean image** within the patch
 - patch variance σ^2 : the average of pixels in the **variance image** within the patch

Plot σ^2 versus μ for all patches.

Problem 2.3: The relationship resemble a line of the form

$$\sigma^2 = a\mu + b$$

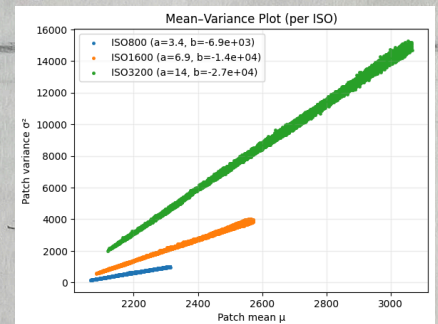
Explain what the **slope** a and **intercept** b represent in terms of noise sources.

a represents the Poisson noise (shot noise)

b represents the read noise

Checkoff

When you're ready, ask a TA to check you off.



3 Simulating Simple Camera Sensor Noise

In this section, you will implement simple numerical models to simulate how noise arises in digital camera sensors. Real images are affected by randomness from photon arrivals and electronic readout circuits. We will model these effects using probability distributions. You will start by simulating photon noise, then read noise, and finally combine them into a full sensor model.

Problem 3.1: If an image pixel has normalized intensity $x \in [0, 1]$, and the maximum possible photon count is $N_{\max}$, then the expected number of detected photons is $\lambda = x \cdot N_{\max}$. The measured photon count follows a Poisson distribution:

$$N \sim \text{Poisson}(\lambda).$$

Write a Python function `add_poisson_noise(img, Nmax)` that simulates photon noise by sampling Poisson-distributed photon counts. Your implementation should preserve the relationship between signal intensity and noise variance. (Hint: Use `np.random.poisson` to generate Poisson-distributed samples.)

Problem 3.2: In addition to photon noise, camera sensors introduce electronic noise during readout. This noise is commonly modeled as additive Gaussian noise with zero mean. We assume that the read noise follows a Gaussian distribution:

$$n \sim \mathcal{N}(0, \sigma^2),$$

where σ is the standard deviation of the read noise.

Write a Python function `add_gaussian_noise(img, sigma)` that adds zero-mean Gaussian noise to an image. (Hint: Use `np.random.normal(mean, std, size)` to generate Gaussian noise.)

Problem 3.3:

Using the functions `add_poisson_noise` and `add_gaussian_noise`, visualize how noise affects an image. Let $x \in [0, 1]$ be a normalized grayscale image. Choose five different values of $N_{\max}$ and σ to generate Poisson-noisy and Gaussian-noisy images with increasing noise levels. Display the original image and the noisy images in two rows, with Poisson noise on the top row and Gaussian noise on the bottom row. Your results should clearly show how increasing noise degrades image quality.

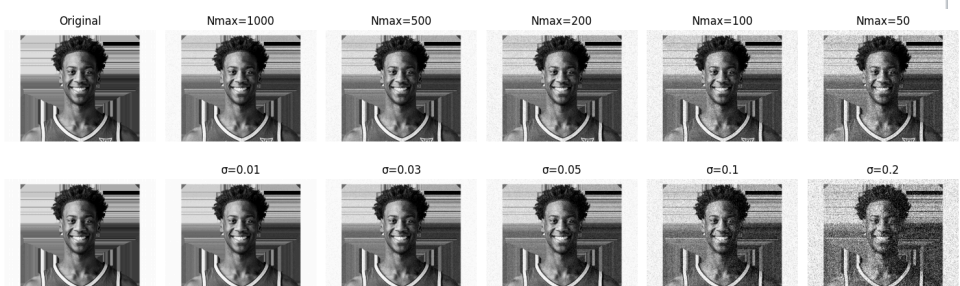
Problem 3.4: A simplified model of a camera sensor combines photon noise and read noise:

$$I = gL + n,$$

where $L \sim \text{Poisson}(\lambda)$ is the photon count, g is the sensor gain, $n \sim \mathcal{N}(0, \sigma^2)$ is the read noise, I is the final measured intensity.

Write a Python function `sensor_model(img, Nmax, g, sigma)` that simulates the full sensor pipeline. (Hint: You can reuse your implementations from Problems 3.1 and 3.2)

Image from 3.3 ->



4 Blind Noise Classification

You are given a clean image X and two noisy observations: Y_1, Y_2 . One image was corrupted by **Gaussian noise only**, and the other by **Poisson noise only**.

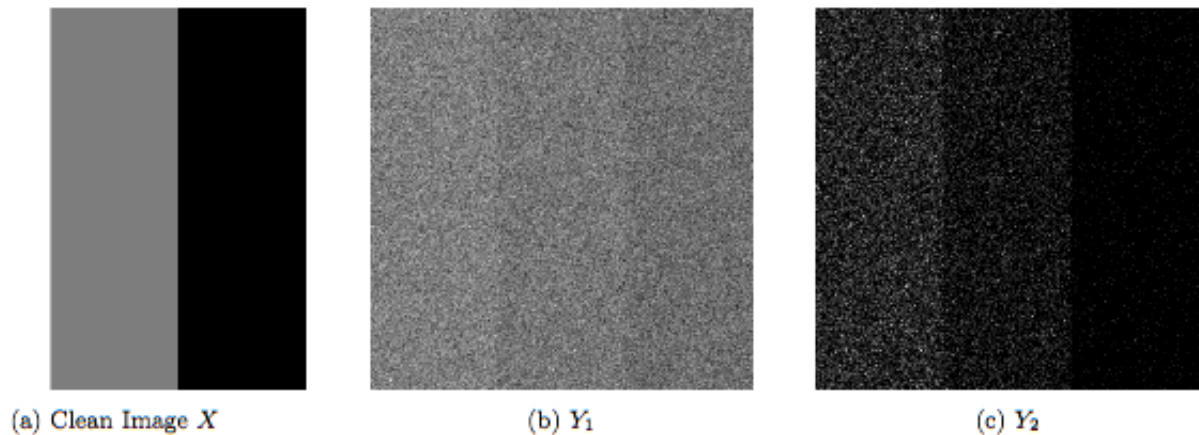


Figure 1: Clean and noisy images

You are also given two mean–variance plots computed from image patches, where each plot shows patch mean (x-axis) versus patch variance (y-axis).

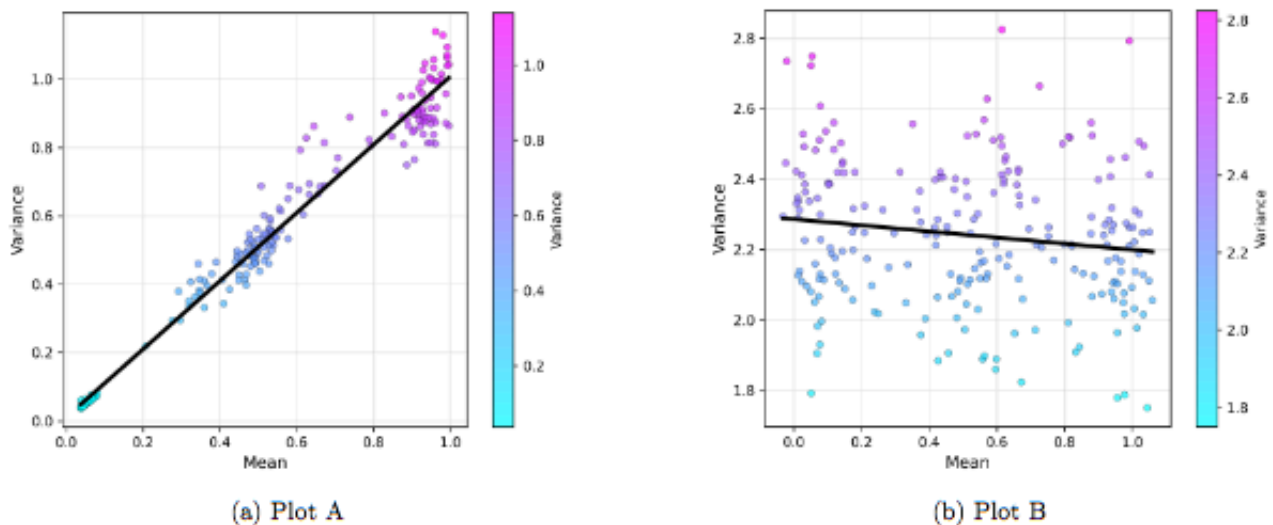


Figure 2: Comparison of Plot A and Plot B

(a) Match each noisy image (Y_1, Y_2) to its corresponding plot (Plot A or Plot B).

Plot A: Y_2

Plot B: Y_1

(b) Determine which noisy image contains Gaussian noise and which contains Poisson noise.

Poisson Noise: Y_2

Gaussian Noise: Y_1

(c) Justify your answer using the statistical relationship between mean and variance.

For a Poisson distribution X , $E[X] = \text{Var}(X)$. This lines up with plot A's linear relationship between mean and variance. For image Y_2 , we see that the noise is more prominent in the lighter portion in the images. This lines up with how the poisson distribution has a greater variance for higher mean (lighter color) patches of the image.

For a Gaussian distribution Z , $\text{Var}(Z) = \sigma^2$ is unrelated to $E[Z]$. This lines up with plot B's near horizontal line, where each patch has an uncorrelated variance relative to mean. For image Y_1 we see that the relative fuzziness looks identical across the image which lines up with the fact that the variance is independent of the mean (brightness) of each patch of the image.

5 Simulating Cheap vs Expensive Cameras

In this problem, you will compare how two cameras perform in low-light conditions by simulating a simple sensor pipeline. You are provided starter code that: (i) creates a synthetic test image `make_test_image()`, (ii) calls a partially-complete sensor model `sensor(...)`, and (iii) evaluates image quality using an SNR-like metric `snr_like(...)`.

You will simulate two cameras with the following parameters:

Camera A (Low-Cost):

- Quantum efficiency: $QE = 0.5$
- Read noise std: $\sigma_{\text{read}} = 6.0$ electrons
- ADC noise std: $\sigma_{\text{ADC}} = 1.2$ DN
- Bit depth: 10 bits

Camera B (High-End):

- Quantum efficiency: $QE = 1.0$
- Read noise std: $\sigma_{\text{read}} = 1.5$ electrons
- ADC noise std: $\sigma_{\text{ADC}} = 0.5$ DN
- Bit depth: 14 bits

Problem 5.1: Complete `TODO 1` in the starter code by filling in `sensor(...)` to simulate:

- Photon noise using a Poisson model,
- Read noise using zero-mean Gaussian noise (in electrons),
- ADC noise using zero-mean Gaussian noise (in DN),
- Clipping/quantization based on the given bit depth.

Problem 5.2: Complete `TODO 2` by implementing `snr_like(Y, mask)` which computes

$$\text{SNR-like} = \frac{\text{mean}}{\text{std}}$$

over a selected region. Use the provided masks (`dark` and `mid`) to report SNR-like values for Camera A and Camera B in:

- a dark region, and
- a mid-tone region.

Dark region SNR:

Camera A: 0.7152811267310355

Camera B: 0.9267500380069466

Mid-tone region SNR:

Camera A: 1.0517586901372156

Camera B: 2.523188286130836

Problem 5.3: Based on your observations, decide which camera performs better in low light and explain your reasoning.

Based on this test, the High-End camera works better than the Low-Cost camera in low light conditions. First off, when looking at the image, the dark portion of the image is much more visible through the High-End camera relative to the Low-Cost. On top of this, the SNR ratio is higher for the High-End camera compared to the Low-Cost camera for the dark region of the image.